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Cylinder Liner Seal Rings (t068)

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Service Parts Topic is to announce the release of new brass seal rings for the K Series cylinder liners. The new seal rings have three locating tabs on the inside diameter. Refer to Figure 1. The tabs will hold the seal ring in the correct position on the cylinder liner during installation of the liner in the cylinder block thus preventing misassembly that can lead to coolant leaks at the liner counterbore.

The tabs are manufactured to have an interference fit to the Lower Press Fit (LPF) diameter of the cylinder liner so the tabs will be positioned in the small radius immediately under the liner flange holding the seal ring in position. Due to the interference fit, during installation of the seal ring it is necessary to keep the seal ring straight on the liner and push with your fingers on the seal ring near the tabs to work it down over the LPF diameter. Do **not** push on the seal rings too hard or the tabs can be deformed to the point the seal ring will **not** fit squarely on the bottom of the liner flange. The thickest seal rings can deform to a very slight angle during installation on the liner. Some deformation is acceptable as long as the tabs will still engage the radius under the liner flange. No sealant or special tools are necessary to install the new seal rings.

Which seal ring to use depends upon which cylinder liner is to be used.

- Table 1 explains the seal rings to be used on K19 cylinder liners that have a **standard** LPF outside diameter.
- Table 2 explains the seal rings to be used on K19 cylinder liners that have an 0.51 mm [0.020 in] **oversize** LPF outside diameter used in a K19 cylinder block that has been salvaged per Service Parts Topic 92T1-18.
- Table 3 explains the seal rings to be used on K38 and K50 cylinder liners whether used in a K38 or K50 cylinder block or a K19 cylinder block that has been salvaged per Service Parts Topic 92T1-18.

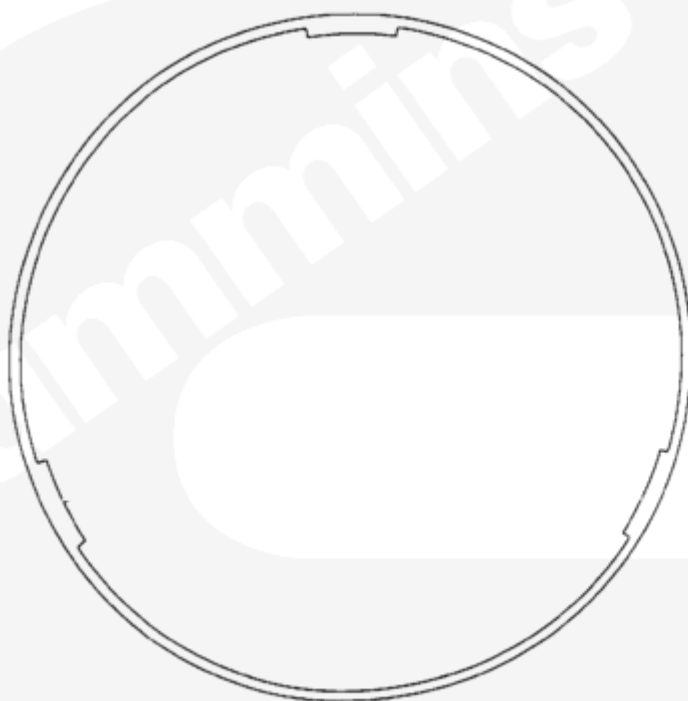
The seal rings with tabs are to be used with the cylinder liners given in Tables 1, 2 and 3 regardless of the style of the cylinder block.

Note : Seal rings without tabs, that are to be used to depletion, must be used per the instructions within Service Parts Topic 92T1-18.

All three tabs on all the seal rings except those used on cylinder liners with the standard K19 LPF outside diameter have notches to aid in identification. The notches show the inside diameter, **not** thickness. The seal rings with tabs are **not** marked to show thickness. It will be necessary to use outside micrometers to determine thickness. The thickness specifications in Table 1, 2, and 3 are the nominal thickness. Thickness tolerance on the 0.52 mm [0.0205 in] thick seal rings is ± 0.020 mm [0.0004 in]. Thickness tolerance on all other seal rings is 0.013 mm [0.0005 in].

Note : This is a product improvement not subject to campaign.

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Figure 1 - K Cylinder Liner Seal Rings with Tabs

TABLE 1: Seal Rings for STANDARD LPF Diameter K19 Cylinder Liners

Liner Kit Part No.	Bare Liner Part No.	Liner UPF/LPF OS/OS	Notes 1	USE TABBED SEAL RING			
				Part No.	Nominal Thickness		Number
					mm	in	Notches
3028434	3202240	STD/STD	Production	3088378	0.46	0.018	None
3028430	3202241	20/STD		3088298	0.52	0.0205	None
3028431	3202242	40/STD		3088379	0.56	0.022	None
3028432	3074552	60/STD		3088380	0.79	0.031	None
3028433	3053800	95/STD					
3009549	3007710	STD/STD	2				

TABLE 2: Seal Rings for OVERSIZE LPF Diameter K19 Cylinder Liners							
Liner Kit Part No.	Bare Liner Part No.	Liner UPF/LPF OS/OS	Notes 1	USE TABBED SEAL RING			
				Part No.	Nominal Thickness		Number
					mm	in	Notches
3803505	3074547	20/20		3088381	0.46	0.018	One
3803509	3074551	60/20		3088382	0.52	0.0205	One
				3088383	0.56	0.022	One
				3088384	0.79	0.031	One

TABLE 3: Seal Rings for STANDARD and OVERSIZE UPF K38 and K50 Cylinder Liners							
Liner Kit Part No.	Bare Liner Part No.	Liner UPF/LPF OS/OS	Notes 3	USE TABBED SEAL RING			
				Part No.	Nominal Thickness		Number
					mm	in	Notches
3007525	3022157	STD/STD	4	3088385	0.46	0.018	Two
3012611	3023017	10/STD	4, 5	3088386	0.52	0.0205	Two
3012612	3023018	20/STD	4, 5	3088387	0.56	0.022	Two
3801268	3023019	30/STD	4, 5	3088388	0.79	0.031	Two
3801269	3023020	40/STD	4, 5				

Notes:

1. Oversize relative to production K19 cylinder liner.
2. Thin flange K19 cylinder liner.
3. The LPF diameter on the K38 and K50 cylinder liners is 1.60 mm [0.063 in] oversize than the LPF diameter on the K19 production cylinder liners.
4. K38 and K50 cylinder liner.
5. Cylinder liners **not** recommended for use in a K19 cylinder block.

Document History

Date	Details
2011-2-14	Module Created
2012-10-29	QSOL Quick Fix Reason: Incorrect Part Number Notes: none

